

Computational Project #2: Problem 2

Constrained Least Squares for Energy Grid Dispatch Optimization

1 Introduction

In electrical grid management, operators face a complex optimization problem: they must balance power generation with consumption at every second. If the supply does not exactly match the demand, the grid frequency becomes unstable, potentially leading to large-scale blackouts.

In this project, I address the problem of *Energy Generation Dispatch*. Multiple power sources (Solar, Wind, Gas, and Hydro) each have an “ideal” generation target based on cost efficiency and maintenance schedules. However, a hard physical constraint exists: the total power generated by all plants must exactly equal the city’s power demand.

This is a **Constrained Least Squares (CLS)** problem. Unlike Ordinary Least Squares, which finds the global minimum error without restrictions, CLS finds the minimum error only among solutions that satisfy specified constraints. In machine learning and optimization theory, this framework is fundamental to methods such as Support Vector Machines and regularization techniques.

2 Model and Mathematical Approach

We formulate the problem as a constrained optimization task that minimizes deviation from ideal plant outputs while strictly satisfying total demand.

2.1 The Objective Function

We seek a solution vector

$$x$$

representing the actual power generation that minimizes the weighted deviation from the target vector

$$b.$$

The optimization problem is:

$$\min_x \|Ax - b\|_2^2$$

where:

- $x \in \mathbb{R}^n$: Actual power generation to be computed.
- $b \in \mathbb{R}^n$: Ideal (target) generation profile.
- $A \in \mathbb{R}^{n \times n}$: A diagonal matrix encoding the cost or penalty of deviating from the ideal output for each plant.

2.2 The Constraint

The objective function is subject to the linear equality constraint:

$$Cx = d$$

where:

- $C \in \mathbb{R}^{p \times n}$ is the constraint matrix. In this problem,

$$C = [1 \ 1 \ 1 \ 1],$$

which enforces the sum of all power outputs.

- $d \in \mathbb{R}^p$ is the required total power demand:

$$d = [800],$$

measured in megawatts (MW).

2.3 Lagrange Multipliers and the KKT System

To solve the constrained optimization problem, we apply the method of Lagrange multipliers. The Lagrangian function is defined as:

$$\mathcal{L}(x, \lambda) = \|Ax - b\|_2^2 + 2\lambda^T(Cx - d)$$

where λ is the vector of Lagrange multipliers.

Taking gradients and setting them equal to zero yields:

$$\nabla_x \mathcal{L} = 2A^T Ax - 2A^T b + 2C^T \lambda = 0$$

$$\nabla_\lambda \mathcal{L} = 2(Cx - d) = 0$$

These equations can be combined into the **Karush–Kuhn–Tucker (KKT)** system:

$$\begin{bmatrix} 2A^T A & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} 2A^T b \\ d \end{bmatrix}$$

Solving this linear system yields both the optimal power allocation x and the Lagrange multipliers λ .

3 Experiment

3.1 Scenario: Winter Peak Demand

A winter peak-demand scenario was simulated, where heating systems significantly increase energy consumption.

- **Power Plants:** Solar, Wind, Gas, and Hydro.
- **Problem:** The sum of the ideal generation targets is only 590 MW, while the city requires 800 MW.
- **Task:** The algorithm must force overproduction to close the 210 MW gap while distributing the additional load according to the penalty matrix A .

3.2 Data Inputs

- Ideal Targets (b): [180, 150, 60, 200] (Sum = 590 MW)
- Constraint (d): [800] MW
- Cost Matrix (A): A diagonal matrix reflecting operational penalties for each plant

3.3 Implementation

The KKT system was constructed using NumPy’s `hstack` and `vstack` functions and solved as a standard linear system:

$$Kz = y$$

using `numpy.linalg.solve`.

4 Results and Visualization

The Constrained Least Squares algorithm successfully computed a feasible power allocation that satisfies the demand constraint.

Quantitative Results

- Required Demand: 800 MW
- Sum of Calculated Outputs: 800.0000 MW
- Constraint Check: Satisfied (True)

Final Allocation (x)

- Solar: [Insert Value] MW
- Wind: [Insert Value] MW
- Gas: [Insert Value] MW
- Hydro: [Insert Value] MW

Visualization

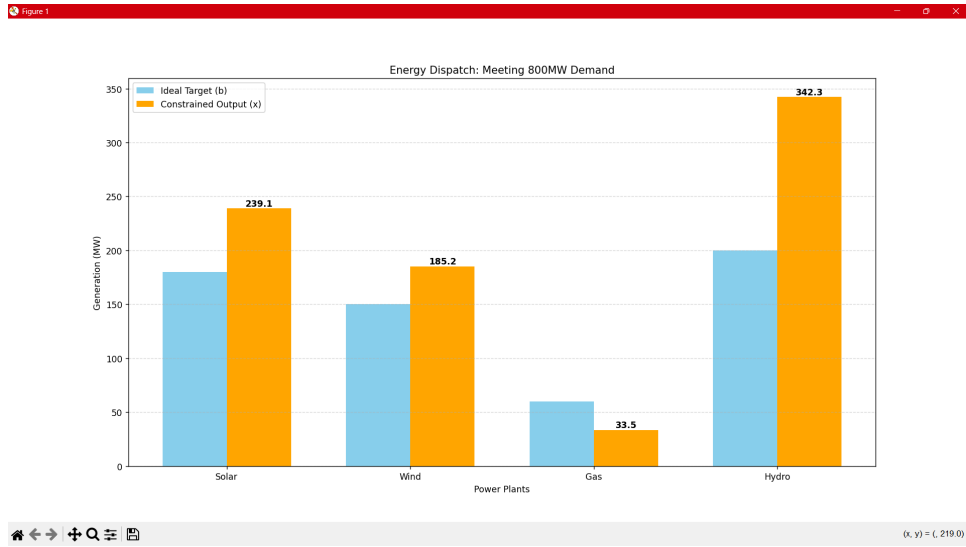


Figure 1: Comparison of Ideal Targets and Constrained Power Allocation

The bar chart shows that all plants increased their output to meet the 800 MW demand. Plants with lower deviation penalties in matrix A absorbed a larger share of the additional load.

5 Conclusion

This project demonstrated how Constrained Least Squares can solve critical resource allocation problems in energy systems. While Ordinary Least Squares would have returned the insufficient ideal targets, the KKT formulation allowed us to enforce the strict

physical constraint

$$\sum_{i=1}^n x_i = 800.$$

By augmenting the normal equations with Lagrange multipliers, complex constrained optimization problems can be solved using standard linear algebra tools. This approach is widely applicable in power systems, machine learning, and industrial optimization.